

An ultrasensitive photoelectrochemical platform for quantifying photoinduced electron-transfer properties of a single entity

Wei Ma^{ID}, Hui Ma, Yue-Yi Peng, He Tian^{ID}* and Yi-Tao Long*

Understanding the photoinduced electron-transfer process is of paramount importance for realizing efficient solar energy conversion. It is rather difficult to clarify the link between the specific properties and the photoelectrochemical performance of an individual component in an ensemble system because data are usually presented as averages because of interplay of the heterogeneity of the bulk system. Here, we report a step-by-step protocol to fabricate an ultrasensitive photoelectrochemical platform for real-time detection of the intrinsic photoelectrochemical behaviors of a single entity with picoampere and sub-millisecond sensitivity. Using a micron-thickness nanoparticulate TiO₂-filmed Au ultramicro-electrode (UME) as the electron-transport electrode, photocurrent transients can be observed for each individual dye-tagged oxide semiconductor nanoparticle collision associated with a single-entity photoelectrochemical reaction. This protocol allows researchers to obtain high-resolution photocurrent signals to quantify the photoinduced electron-transfer properties of an individual entity, as well as to precisely process the data obtained. We also include procedures for dynamic light scattering (DLS) analysis, transmission electron microscopy (TEM) imaging and collision frequency-concentration correlation to confirm that the photoelectrochemical collision events occur at an unambiguously single-entity level. The time required for the entire protocol is ~36 h, with a single-entity photoelectrochemical measurement taking <1 h to complete for each independent experiment. This protocol requires basic nanoelectrochemistry and nanotechnology skills, as well as an intermediate-level understanding of photoelectrochemistry.

Introduction

Photoelectrochemical reactions have gained much attention among researchers over the past decades, mainly because of their powerful utility in solar energy conversion^{1–3}. Photoinduced electron transfer is an essential step in the conversion of light into electric power in photoelectrochemistry^{4,5}, also frequently called photo-driven electrochemical reactions by chemists^{6,7}. Among the most efficient photoelectrochemical reactions are those involving light absorption by wide-band-gap semiconductor nanoparticles and molecular adsorbates, which result in the occurrence of electron-transfer reactions at the semiconductor interface that are the basis for photoelectric conversion in photovoltaic devices^{2,8,9}. Therefore, understanding the photoinduced electron-transfer process is of paramount importance for realizing efficient conversion of solar energy. A range of analytical methods can be used to study photoelectrochemical properties, such as time-resolved photocurrent measurement, photovoltage measurement and electrochemical impedance spectroscopy^{4–8}. These techniques are capable of investigating the physicochemical properties and photoelectric performance of different types of photoelectrochemical processes; however, substantial enhancements of the design of highly efficient photoelectrochemical devices remain a great challenge for the following reasons^{10,11}. The inability to quantify the structure–activity relationship because of the interplay of structural and functional heterogeneity among individual entities is a major barrier^{12,13}. The intrinsic properties are masked in traditional photoelectrochemical measurements, producing averaged data caused by measurement of large ensembles of molecules or billions of nanoparticles. This could result in overlooking a subpopulation minority species with outstanding activities^{14–17}. In addition, the surrounding medium and assembly process are important factors that affect conversion efficiencies. Therefore, thorough insights into the photoelectrochemical performance of a single entity would

Key Laboratory of Advanced Materials, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai, People's Republic of China. *e-mail: tianhe@ecust.edu.cn; yitlong@ecust.edu.cn

allow the roles of nanoparticles and dye molecules to be quantified, uncovering the intrinsic properties of photoinduced transfer under realistic reaction conditions.

In contrast to studies on a bulk system, single-entity electrochemistry has emerged as a unique tool for exploring the structural and kinetic features of individual entities during stochastic collision measurements at a UME surface, avoiding any inhomogeneous averaging and broadening effects^{18–20}. Many examples of single-entity collision experiments have been reported in this field, such as electrocatalysis^{21–23}, blocking^{24,25} and electrochemical stripping^{26–29}. For example, the electrocatalytic behavior of individual entities has been studied as they collide on a relatively inert UME, resulting in current transients due to the catalytic amplification-mediated electron-transfer process^{19–23}. Quantitative information about individual nanoparticles, such as size, morphology and concentration, can be obtained by measuring the faradic charge associated with the electrochemically destructive process of direct oxidation of each colliding nanoparticle^{26,27}. Recently, this technique has also been utilized to investigate the electron-transfer dynamics of individual nanoparticles at the UME interface^{30–33}. Therefore, stochastic collision electrochemical measurements offer direct electron-transfer information at interfaces and provide valuable evidence in studying the intrinsic electrochemical activities of an individual entity^{18–30}. In addition, they also have advantages that include rapid and sensitive measurement, low cost, lack of requirement for labeling, high throughput and low sample consumption^{31–35}. If we could use single-entity electrochemistry to study the photoelectrochemical reaction, we could solve the ensemble problem described above. However, the key bottleneck in applying single-entity electrochemistry to photoelectrochemical measurement is the need for real-time recording of the ultrafast photoinduced electron-transfer data³⁶. That is, the electron-transfer process of a photoelectrochemical reaction occurs on a sub-picosecond timescale and cannot be electrochemically resolved because of the limitations of current electrochemical instruments. Current photoinduced electron-transfer research is mainly based on ultrafast time-resolved transient spectroscopy techniques for tracking the intermediate states of the electron-transfer process in photoelectrochemical reactions on picosecond and femtosecond timescales^{37–39}.

Development and applications of the protocol

A useful strategy for achieving in situ electrochemical measurement of photoinduced electron transfer at the single-entity level is to fabricate a micron-thickness nanoparticulate TiO₂ film-coated Au UME (TiO₂@Au UME) as an electron-transport electrode. At this electrode, photoinduced electrons undergo a trap-limited diffusion process in the TiO₂ film, causing a substantial extension of electron transit time. Photoinduced electrons move through the TiO₂ film to the Au UME electrode in a multiple-trapping process. Upon visible light absorption by N719 dye molecules, the incident photons are converted into electron-hole pairs, and a gradient electrical potential is frequently generated to cause the separation of electron and hole. Photoexcited electrons are subsequently injected from the excited state of N719 into the semiconductor conduction band of neighboring ZnO or TiO₂ nanoparticles. When individual N719@TiO₂ or N719@ZnO entities stochastically collide with TiO₂@Au UME, the electrons are transferred into the nanoparticulate TiO₂ film. The electrons then travel through the TiO₂ film to reach the Au UME electrode. The photoexcitation, electron separation and injection are usually ultrafast and take place on a picosecond timescale. Therefore, in our photoelectrochemical measurements, the rate-determining steps are the collision of a nanoparticle with the TiO₂@Au UME and/or the electron diffusion process inside the TiO₂ film.

To elucidate the electron-transfer dynamics of the collision process, we studied the photoelectrochemical behaviors of individual dye-tagged oxide semiconductor nanoparticles colliding with a bare Au UME. No current signal was observed under illumination in this case (Supplementary Fig. 1). This demonstrates that the collision occurs in a very short time at bare Au electrode surfaces and is faster than the acquisition time of a single-point electrochemical measurement. In this study, we performed single-entity photoelectrochemical experiments using a low-noise electrochemical instrument at a sampling rate of 100 kHz. The internal low-pass Bessel filter of the amplifier was set to 5 kHz. This means the collision process takes place in <66 μ s, as estimated by instrumental resolution. To capture the transient signal, we needed to substantially extend the residence time of the nanoparticle on the electrode or greatly slow the electron transfer in the TiO₂@Au UME. Therefore, we designed a micron-thickness semiconductor film of nanoparticulate TiO₂ and coated the surface of an Au UME with it so that photoinjected electrons proceed with trap-limited diffusion in the TiO₂ film. As a result, the typical duration of electron transit in TiO₂ film was considerably increased, allowing the achievement of successful real-time detection of single-entity photoelectrochemical

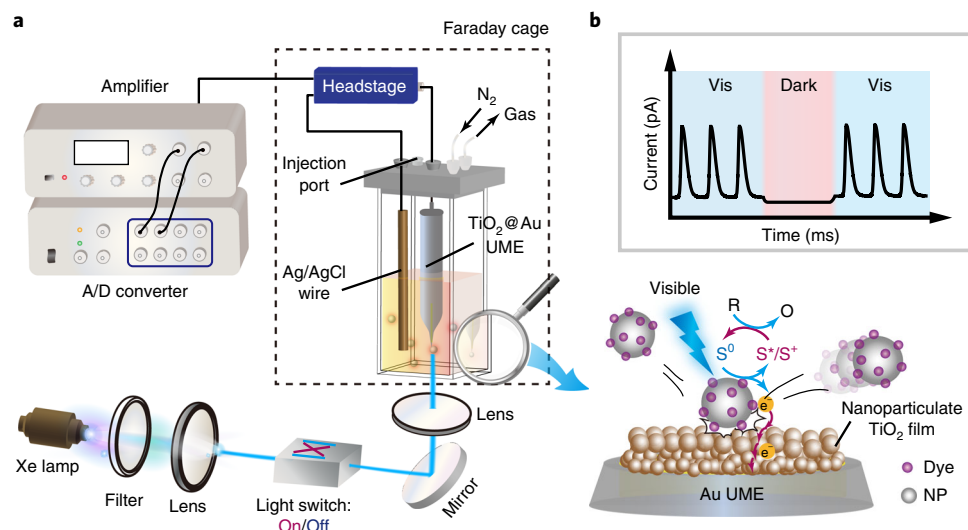


Fig. 1 | Single-entity photoelectrochemical measurement. **a**, Schematic illustration of an ultrasensitive photoelectrochemical platform. **b**, Representative photoelectrochemical responses for individual dye-tagged oxide semiconductor nanoparticles during a collision at a $\text{TiO}_2\text{@Au}$ UME under visible light ($\lambda > 450\text{ nm}$, Xe lamp). When visible light shines, dye molecules are excited from their ground state (S^0) to a higher energy state (S^*). The oxidized dye molecule (S^+) is again regenerated by electron donation from the reduced species (R) in the electrolyte, generating the oxidized species (O). A/D, analog-to-digital; NP, nanoparticle; Vis, visible light.

behaviors. Owing to a trap-mediated electron diffusion process in TiO_2 films, we observed sub-millisecond photocurrent transients associated with single-entity photoelectrochemical reactions.

Using this approach, we developed an ultrasensitive photoelectrochemical platform for real-time photocurrent detection of individual dye-tagged oxide semiconductor nanoparticles with picoampere and sub-millisecond sensitivity^{40,41} (Fig. 1). In particular, the high resolution and sensitivity of this technique could enable a wide range of potential applications for single-entity photoelectrochemical measurements, for example, in dye-sensitized solar cells, photoelectrochemical water splitting and photocatalysis systems.

This strategy greatly enhances the possibilities of applying single-entity electrochemistry to a broad range of photoelectrochemical measurements. Based on high-resolution photoelectrochemical measurements, our platform can be used for quantifying the photoinduced electron-transfer process of a single event, enabling electron-transfer dynamics analysis at a semiconducting TiO_2 film. This may have considerable impact in the field of intrinsic performance of photoelectrochemical devices. It will certainly help to achieve an improved understanding of the electron-transfer process at the semiconductor interface, where the photoelectrochemical reaction processes take place.

Limitations and alternative methods

Owing to the use of $\text{TiO}_2\text{@Au}$ UME as an electron-transport electrode, an energy level match between dye/metal oxide nanoparticles and the TiO_2 semiconductor interface is critical for efficient electron collection⁴². To obtain high efficiency for injection of dye molecules into the semiconductor, a good overlap between the excited-state levels of the dye molecule and the conduction band levels of the semiconductor is required⁴³. Among the different types of dyes, ruthenium(II) (Ru(II))-based systems—such as N719, N3, and black dye—are the most popular for use with the liquid iodide/triiodide (I^-/I_3^-) electrolyte^{44,45} and zinc(II) porphyrins have reached high efficiency when coupling to cobalt(II)/(III) redox shuttles⁴⁶. Moreover, a number of metal-free dyes⁴⁷ can also be used in our photoelectrochemical measurement system. However, not all metal oxide nanoparticles and dyes will work with $\text{TiO}_2\text{@Au}$ UME because of mismatched energy levels. In addition, considering the extreme sensitivity of individual event signals, testing the effects of the filter and amplifier of a low-noise electrochemical instrument should be performed as a ‘best practice’ standard in single-entity photoelectrochemical studies to minimize the electronics’ effects on measurement results^{48,49}. Such effects can potentially limit the general applicability of this method. To assist researchers in choosing the most suitable preparation approach for metal oxide-filmed UMEs, alternative methods^{50–54}, including their relative advantages and limitations, are outlined in Supplementary Table 1.

The approach described here also has some other limitations. (i) Nanoparticle suspensions should consistently remain monodispersed during the entire photoelectrochemical measurement, which can be achieved by controlling the concentrations of nanoparticle suspensions and the electrolyte solution at the (approximately) picomolar and millimolar levels, respectively. (ii) The applied potential should remain in the range from +800 to −600 mV versus Ag/AgCl to prevent any damage to the TiO₂ film on the Au UME. (iii) Incident light intensity should be <200 mW. Higher light intensity causes an increase in current background noise due to local thermal drift.

In this protocol, we describe step-by-step procedures for the fabrication of TiO₂@Au UMEs, the preparation of monodispersed dye-tagged oxide semiconductor nanoparticle suspensions and the subsequent procedures to monitor the photoelectrochemical behaviors of individual entities in real time, process data and confirm single-nanoparticle events. As a model system, we use the N719 dye molecules²⁷ and TiO₂/ZnO semiconductor nanoparticles⁵ as the photosensitizers and charge transport carriers, respectively. As demonstrated, this platform offers picoampere and sub-millisecond sensitivity in terms of the current magnitude and duration of individual collision events at a TiO₂@Au UME interface, allowing the quantitative measurement of the photoinduced electron-transfer processes of a single entity. We strongly believe this protocol could pave the way for the fabrication of high-performance photoelectrochemical devices by quantifying the contributions of individual components with single-entity photoelectrochemical measurements.

Materials

Reagents

- Di-tetrabutylammonium cis-bis(isothiocyanato)bis(2, 2'-bipyridyl-4, 4'-dicarboxylato)ruthenium(II) (N719; Sigma-Aldrich, cat. no. 703214)
- Titanium dioxide nanoparticles (99.7% (wt/wt), <25-nm particle size; Sigma-Aldrich, cat. no. 637254)
- Acetonitrile (≥99.9% (wt/wt), CH₃CN; Sigma-Aldrich, cat. no. 34851)
- Lithium iodide (99.9% (wt/wt), LiI; Sigma-Aldrich, cat. no. 518018)
- Iodine (≥99.99% (wt/wt), I₂; Sigma-Aldrich, cat. no. 326143)
- Lithium perchlorate (99.99% (wt/wt), LiClO₄; Sigma-Aldrich, cat. no. 634565)
- Methanol (≥99.9% (wt/wt), CH₃OH; Sigma-Aldrich, cat. no. 34860) **!CAUTION** Methanol is flammable and toxic and should be handled in a fume hood.
- Ethanol (95% (wt/wt), CH₃CH₂OH; Sigma-Aldrich, cat. no. E714-8)
- Titanium(IV) chloride solution 1.0 M in methylene chloride (TiCl₄; Sigma-Aldrich, cat. no. 249866) **!CAUTION** TiCl₄ is toxic and corrosive. Use gloves, goggles and other chemical-use safety measures that are recommended by the appropriate concerned agencies.
- Acetone (≥99.8% (wt/wt), CH₃COCH₃; Sigma-Aldrich, cat. no. 34850) **!CAUTION** Acetone is acutely toxic and it is an unstable explosive. Avoid flames and handle it under a fume hood. Dispose of acetone in accordance with relevant local regulations.
- Potassium hydroxide (KOH; Aladdin, cat. no. P112281) **!CAUTION** KOH causes skin corrosion and can cause serious damage to the eyes.
- Dimethyl sulfoxide (≥99.7% (wt/wt), DMSO; Sigma-Aldrich, cat. no. 34869)
- Ferrocene (98% (wt/wt); Sigma-Aldrich, cat. no. F408)
- Tetrabutylammonium hexafluorophosphate (≥99.0% (wt/wt), TEAPF₆; Sigma-Aldrich, cat. no. 86879)
- Tetrabutylammonium perchlorate (98% (wt/wt), TBAP; Alfa Aesar, cat. no. C42014)
- Chloroform (CHCl₃; Sigma-Aldrich, cat. no. C2432) **!CAUTION** CHCl₃ is flammable and toxic, and should be handled in a fume hood.
- Dichloromethane (CH₂Cl₂; Sigma-Aldrich, cat. no. D65100) **!CAUTION** CH₂Cl₂ is a possible carcinogen and is toxic upon inhalation; it causes irritation and burning pain upon prolonged contact.
- Water (Milli-Q, 18.2 MΩ·cm at 25 °C)
- High-purity nitrogen (99.998% (wt/wt), N₂; Cryogenic Gases, cat. no. UN1066) **!CAUTION** High-purity nitrogen poses an asphyxiation risk. It should be handled in a well-ventilated space.
- High-purity argon (99.998% (wt/wt); Ar; Cryogenic Gases, cat. no. UN1006)
- TiO₂ paste (18NR-T; hereafter referred to as 'paste A'; Greatcell Solar, cat. no. 0002010)
- TiO₂ paste (18NR-AO; hereafter referred to as 'paste B', mixture with 28.6% (wt/wt) 400-nm TiO₂, 2.9% (wt/wt) 10-nm TiO₂ and 7.2% (wt/wt) ethyl cellulose; Greatcell Solar, cat. no. 002270)
- Zinc acetate (99.99% (wt/wt), Zn(OAc)₂·2H₂O; Sigma-Aldrich, cat. no. 383317)
- 5-Bromovaleric acid (97% (wt/wt), (BrCH₂(CH₂)₂CH₂COOH; Sigma-Aldrich, cat. no. 158410)
- Dimethylformamide (DMF; Alfa Aesar, cat. no. D131-1)

- Hydrogen peroxide (H_2O_2 ; Sigma-Aldrich, cat. no. 323381) **!CAUTION** H_2O_2 can be corrosive and can cause burns; wear personal protective equipment and work in a fume hood when handling it.
- Sulfuric acid (H_2SO_4 ; Sigma-Aldrich, cat. no. 339741) **!CAUTION** H_2SO_4 is extremely corrosive and can cause serious burns; wear personal protective equipment and work in a fume hood when handling it.
- Hydrochloric acid solution (HCl; Sigma-Aldrich, cat. no. 30721) **!CAUTION** HCl solution is corrosive and can cause serious burns; wear personal protective equipment and work in a fume hood when handling it.

Equipment

- Platinum wire counter electrode (CH Instruments, part no. CHI115)
- Ag/AgCl reference electrode (CH Instruments, part no. CHI111)
- Cell stand with a 25 mm $\times$ 40-mm glass cell (CH Instruments, part no. CHI220)
- Gold (Au) wires (10- μm diameter, 99.99% (wt/wt); Goodfellow, cat. no. AU005110)
- Au foil electrode (99.99% (wt/wt); Goodfellow, cat. no. AU000348)
- Abrasive discs (3.0, 1.0 and 0.3 μm ; Buehler, cat. nos. 693102, 693103 and 693104)
- 3M diamond lapping films (3.0, 1.0 and 0.5 μm ; TED Pella, cat. nos. 816-326, 816-328 and 816-330)
- Carbon double-sided tape (8 mm $\times$ 20 m, Nisshin EM, cat. no. 7311)
- Copper double-sided tape (8 mm $\times$ 20 m, Nisshin EM, cat. no. 733)
- Grinding stone (2-cm radius; Electron Microscopy China, cat. no. DZ16328)
- Alumina N, Super 1 (MP Biomedicals, cat. no. 0204580)
- Borosilicate glass capillary (Sutter Instrument, cat. no. B200-116-10)
- Fluorine-doped tin oxide-coated glass slides (FTO; Hartford Glass, model no. TEC 7)
- Silver (Ag) wires (1.0-mm diameter, 99.999% (wt/wt); Alfa Aesar, cat. no. 11466)
- Hydrophobic polytetrafluoroethylene (PTFE) membrane filter (0.22 μm ; RephiLe Bioscience, cat. no. RJF1322NH)
- Micropipette puller (NARISHIGE, model no. PC-10)
- Xenon lamp (500 W; Ocean Optic, model no. XBO 500W/H OFR)
- Si detector (Newport, model no. 71640)
- Vortex mixer (Vision Scientific, model no. BV1000)
- Axygen pipette tips (Axygen Scientific, cat. no. T-200-Y)
- Axygen microcentrifuge tubes (Axygen Scientific, cat. no. MCT-500-C)
- Eppendorf Reference 2 pipettes (single-channel adjustable-volume pipettes; Eppendorf, cat. no. 492000904)
- Ultrasonic cleaner (Shanghai Ultrasonic Instrument, model no. KUDOS SK3200)
- Analytical balance (Mettler Toledo, cat. no. ME104E)
- Millipore Milli Q A10 water purification system (MilliporeSigma, model no. SIMS00000)
- Amplifier (Molecular Devices, model no. Axopatch 200B)
- Low-noise data acquisition system (Molecular Devices, model no. DigiData 1550A)
- Faraday cage (constructed in-house: 50 $\times$ 50 $\times$ 50 cm; Thorlabs, cat. no. PSY406 (for copper mesh); see <https://backyardbrains.com/experiments/faraday> for instructions for making a similar Faraday cage)
- Electrochemical workstation (CH Instruments, model no. CHI660E)
- Vibration isolation table (Shanghai/Taicang Tianhe Electromechanical, cat. no. HAP-100)
- Vinyl electrical tape (Minnesota Mining and Manufacturing, cat. no. 471)
- Aluminum foil (Sangon Biotech, cat. no. F504332)
- Centrifuge (model no. 5430 with rotor FA-45-30-11, including rotor lid; Eppendorf, cat. no. 5427000291)
- Headstage holder system (AS ONE, cat. no. 6-776-03)
- Electric welding machine (Yida Electronic, model no. A-BF936)
- Copper wire (1-mm diameter, 99.9% (wt/wt); Alfa Aesar, cat. no. 010782)
- Fully automatic AC voltage regulator (Chint Electric, model no. TND1(SVC)-5)
- Paintbrush (sable, size 05; Jiangxi Province Mo Pass Culture Development, cat. no. H985)
- Tube furnace (Hefei Ke Jing Materials Technology, model no. OTF-1200X-S)
- Drying oven (Shanghai Yiheng Instruments, model no. DHG-9070(A))
- Optical long-pass filters (Thorlabs, model nos. FEL0400 and FEL0450)
- Optical system (constructed in-house: plano-convex lenses (K9 $d = 50$ mm, $f = 63$ mm, t_c (thickness of the center) = 14 mm, t_e (thickness of the edge) = 3 mm and $d = 50$ mm, $f = 80$ mm, $t_e = 3$ mm,

$t_c = 11.4$ mm; aluminum film reflector $d = 50$ mm; lens and reflector support; Fredor, see <http://www.fredorscience.com/html/product/optical/20131226/56.html> for information on designing an optical system)

- Quartz photoelectrochemical cell (constructed in-house; Wuhan Gaoss Union CnTech, see http://www.gaossunion.com/Products_T51.html for information on making a photoelectrochemical cell)
- DLS Zetasizer Nano-ZS Instrument (Malvern Instruments, model no. ZEM4228)
- Transmission electron microscope (TEM; JEOL, model no. JEM-2100)
- Field emission scanning electron microscope (FE-SEM; Hitachi, model no. Hitachi S-4800)
- UV-visible spectrometer (Ocean Optics, model no. USB 2000+)
- Fourier-transform infrared spectrophotometer (FTIR; Thermo Fisher Scientific, model no. Nicolet-6700)
- X-ray powder diffractometer (XRD; Rigaku, model no. D/MAX 2550 VB/PC)
- Glass vial (AS ONE, cat. no. 8397890)
- Alcohol burner (AS ONE, model no. 8611041)
- Resistor coil (Narishige, cat. no. PC-10)
- Vacuum pump (Sciencetool International Group, model no. T230)
- Three-way stopcock (AS ONE, cat. no. 10D5636)
- Soft rubber tube (AS ONE, cat. no. 02029878)
- Beveler (Narishige, model no. EG-401)
- Glass container (AS ONE, cat. no. 8612301)
- Disposable cuvette (Malvern, model no. DTS0012)
- Glass bottle (AS ONE, cat. no. 8399530)
- Dry cabinet (Eirmai, $38 \times 27 \times 28$ cm, model no. MRD-28)
- Teflon tubing (Aladdin, cat. no. P2790)
- Benchtop sputter-coating system (Hitachi, model no. MC1000)
- Light chopper (Sutter Instrument, model no. LB-SC)

Software

- Custom-designed data analysis software (see Supplementary Software)
- Origin 9.0 (OriginLab: <https://www.originlab.com/index.aspx?go=Support&pid=2051>)
- Zetasizer software pss0012-37 version 7.11 (http://en.findeen.com/malvern_instruments_zetasizer_software.html)

Reagent setup

▲ **CRITICAL** All aqueous solutions are prepared with Milli-Q purified water (18.2 M Ω -cm at 25 °C).

▲ **CRITICAL** All organic solvents are dried by adding activated alumina.

LiClO₄ solution

LiClO₄ solution is 0.1 M LiClO₄ in dry acetonitrile. This electrolyte can be stored at room temperature (25 °C) for 1 week.

Ferrocene solution

Ferrocene electrolyte solution is 1 mM ferrocene and 0.1 M TEAPF₆ in dry acetonitrile.

▲ **CRITICAL** Make fresh solution before each experiment.

Iodide/triiodide (I^-/I_3^-) electrolyte

Iodide/triiodide (I^-/I_3^-) redox electrolyte is 10 mM LiI and 1 mM I₂ in dry acetonitrile. This electrolyte solution can be stored in a brown bottle at room temperature for 1 week.

DMSO–water solution

DMSO–water solution is 2.2 M water and 50 mM TBAP in DMSO solvent. This electrolyte solution can be stored at room temperature for 1 week.

DMSO–methanol solution

DMSO–methanol solution is 2.2 M methanol and 50 mM TBAP in DMSO. This electrolyte solution can be stored at room temperature for 1 week.

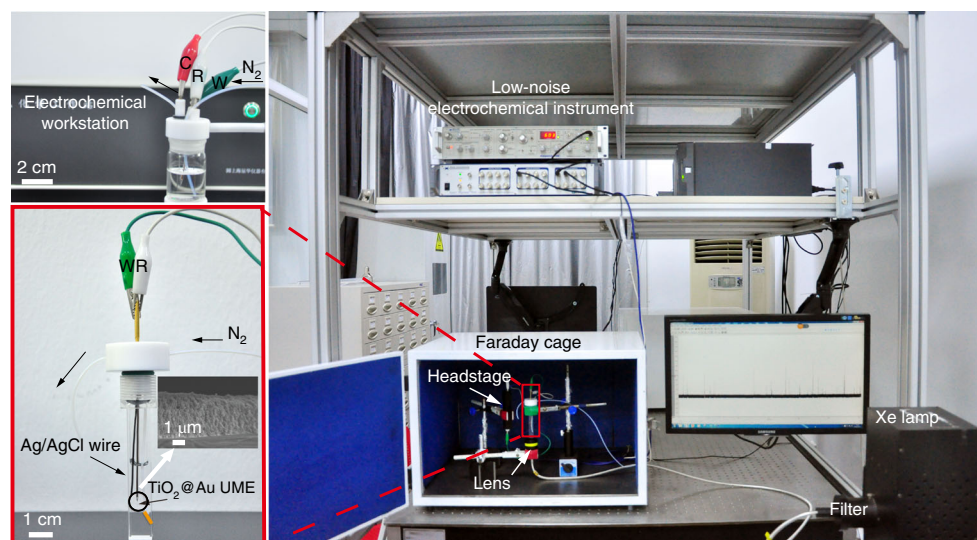


Fig. 2 | Equipment setups for electrochemical measurement and single-entity photoelectrochemical measurement. Top left and right, photographs of an electrochemical cell equipped with three electrodes: a $\text{TiO}_2\text{@Au}$ UME working electrode (W), a platinum wire counter electrode (C) and an Ag/AgCl wire reference electrode (R; top left); and a photoelectrochemical setup for single-entity photoelectrochemical measurement (right). Bottom left, Photoelectrochemical cell equipped with a $\text{TiO}_2\text{@Au}$ UME working electrode (W) and an Ag/AgCl wire reference electrode (R). Inset: SEM image of a nanoparticulate TiO_2 film on an Au UME.

Piranha solution

To make piranha solution, mix 30% (wt/wt) H_2O_2 and 98% (wt/wt) H_2SO_4 at a ratio of 3:7 (vol/vol). Mix the solution in an open glass container when needed. **!CAUTION** Mixing of H_2O_2 and H_2SO_4 is highly exothermic and can be explosive, especially if mixed with any organic solvent. Wear personal protective equipment and work in a fume hood. **!CAUTION** Do not add H_2SO_4 to H_2O_2 , as this will cause intense and rapid heating of the resulting mixture, potentially splashing this strongly oxidizing solution. **!CAUTION** Excess solution should be disposed of in an appropriate manner. Piranha solution must be left in an open container in a safe area to cool for several hours before storing for disposal. Once cooled to room temperature, the solution can be transferred to a closed glass container for waste storage.

Equipment setup

Electrochemical cell

All three-electrode electrochemical experiments are performed using a CHI660E electrochemical workstation. A glass electrochemical cell is set up on a cell stand, with three electrodes inserted into the cell: a $\text{TiO}_2\text{@Au}$ UME (10 μm) as the working electrode, a platinum wire counter electrode and an Ag/AgCl reference electrode (Fig. 2, top left). Add 6 mL of ferrocene electrolyte solution to the electrochemical cell and purge the solution with argon for 10 min through Teflon tubing. Maintain the solution under argon atmosphere during subsequent electrochemical measurements.

Photoelectrochemical cell

All single-entity photoelectrochemical measurements are performed using an Axopatch 200B amplifier in combination with a xenon lamp with a filter to cut off the wavelengths of light <450 nm under an argon atmosphere throughout the experiment in a Faraday cage at 25 $^\circ\text{C}$ (Fig. 2, right). An enclosed quartz electrochemical cell is set up on a cell holder, with two electrodes inserted into the cell: a $\text{TiO}_2\text{@Au}$ UME and an Ag/AgCl reference electrode (Fig. 2, bottom left). The $\text{TiO}_2\text{@Au}$ UME is aligned carefully so that the light beam from the xenon lamp passes through the center of the $\text{TiO}_2\text{@Au}$ UME. To maintain the stability of the defined reference potential over prolonged measurement periods and avoid contamination-induced effects from the reference electrode⁵⁵, the Ag/AgCl wire should be placed at a distance of a few millimeters from the $\text{TiO}_2\text{@Au}$ UME electrode and used routinely in a simplified two-electrode format. Add 3 mL of electrolyte solution (I^-/I^{3-} electrolyte, DMSO–water solution or DMSO–methanol solution) to the photoelectrochemical cell. Purge the solution by bubbling argon for 10 min through the Teflon tubing before measurements. Considering the high photocatalytic

activity of ZnO and TiO₂ nanocrystals under ultraviolet light, unless otherwise stated, visible light (a xenon lamp with wavelength >450 nm) is used. After filtering, the spectral composition <450 nm is completely cut off (Supplementary Fig. 2). In addition, the intensity of light in the vicinity of the TiO₂@Au UME should be 100 mW/cm², which is measured by an Si detector.

Procedure

Fabrication and characterization of TiO₂@Au UMEs ● Timing ~24 h

- 1 Clean a borosilicate capillary (length, 10.0 cm; i.d., 1.16 mm; o.d., 2.0 mm) and an Au wire (length, 1.0 cm; diameter, 10 μm) by sonication sequentially in 10 mL of acetone, ethanol and Milli-Q water in a glass vial in a water bath in an ultrasonic cleaner for 30 min in each solution.
- 2 Rinse the borosilicate capillary and Au wire extensively with copious amounts of Milli-Q water. Dry them in a covered glass vial in an oven at 120 °C for 8 h.
- 3 Pull the borosilicate capillary to produce a micropipette with a tip size of dozens of micrometers using a micropipette puller (two-step pulling, heater level no. 1: 63%; no. 2: 78%).

? TROUBLESHOOTING

- 4 Use an alcohol burner to seal the tip of the micropipette for 1 s so that a conical tip is obtained. Check the micropipette tip under a microscope and make sure that the end of the glass tip is completely sealed.

▲ **CRITICAL STEP** Adjust flame temperature and burning time to avoid undesired bending of the micropipette tip.

- 5 Insert a straight 10-μm Au wire into the sealed micropipette without twisting the wire. Gently tap the capillary to position the Au wire in the crack of the cone of the micropipette tip and check it under the microscope.

▲ **CRITICAL STEP** Wear gloves to prevent contaminating the Au wire. Keep the Au wire straight for easier placement into the capillary.

? TROUBLESHOOTING

- 6 Place the microtip vertically at the center of a heated resistor coil and then connect the micropipette to a vacuum pump with a soft rubber tube. Turn on the vacuum pump and slowly actuate a three-way stopcock to create a vacuum in the micropipette over 30 min.

▲ **CRITICAL STEP** A three-way stopcock is used to prevent the displacement of the Au wire due to a sudden pressure change.

- 7 Turn on the micropipette puller. Slowly move the micropipette deeper into the heating coil to ensure proper sealing as heater level gradually increases from 50% to 70% (Narishige PC-10; seal length: 5 mm). Turn off the heater and vacuum. Let the microtip cool for 5 min. Remove the capillary and check the micropipette under a microscope to ensure that it has been sealed with the Au wire inside and without any air bubbles.

▲ **CRITICAL STEP** Select an appropriate heating level and maintain a high vacuum to prevent the tip from bending and air bubbles from forming during the sealing process.

? TROUBLESHOOTING

- 8 Use a beveler to polish the microtip on a diamond lapping film (3.0 μm) until the Au wire is exposed under microscopy. Polish the obtained Au UME on diamond lapping films (1.0 and 0.5 μm) in sequence to obtain a smooth mirror surface. Rinse with ultrapure water and dry the Au UME for the fabrication of TiO₂ film.

▲ **CRITICAL STEP** Keep the Au UME perpendicular to the diamond lapping films during polishing process to obtain a flat, smooth surface. Rinse the Au UME surface with Milli-Q water and dry with nitrogen flow after each polishing cycle to remove the Au and glass granule debris attached to the surface.

- 9 Dip a sable paintbrush (size 05) into TiO₂ paste A (20 nm) and paint the paste over the surface of the Au UME until the surface is uniformly coated. Dry the TiO₂ paste in a vacuum oven at 125 °C for 8 min. Repeat this painting and drying processes to fabricate several TiO₂ paste A layers on the Au UME. Sequentially, brush-paint the Au UME with the TiO₂ paste B (400 nm) and then dry it in a vacuum oven at 125 °C for 8 min.

▲ **CRITICAL STEP** After each layer of paste A or paste B is brush-painted on the surface of the Au UME, keep it in a clean dry cabinet for 3 min to allow the paste to relax and the surface irregularity to reduce before drying. The thickness of the TiO₂ film depends on the number of layers of TiO₂ paste A. Typically, five layers of paste A and one layer of paste B correspond to 4- to 5-μm film thickness.

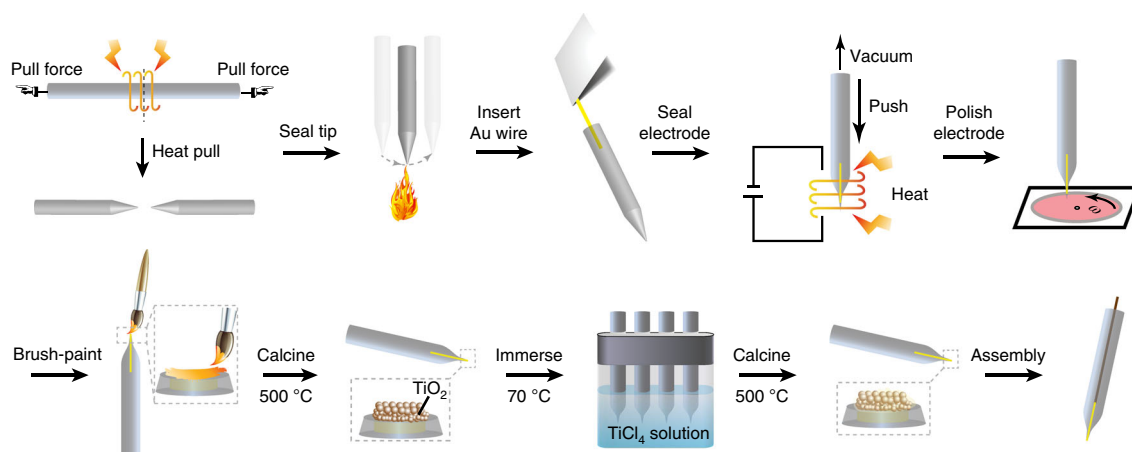


Fig. 3 | Procedure for preparation of TiO_2 @Au UMEs. Pull a borosilicate capillary to make a micropipette and seal the micropipette tip for 1 s. Insert an Au wire into the sealed micropipette and seal the micropipette with Au wire. Polish the Au UME and then brush-paint a TiO_2 layer on an Au UME surface. Calcine the TiO_2 -filmed Au UME and immerse it into a TiCl_4 aqueous solution. Calcine the TiO_2 film again and assemble the TiO_2 @Au UME by inserting a thick copper wire into the capillary to establish an electrical connection with Au wire (see also Supplementary Video 1).

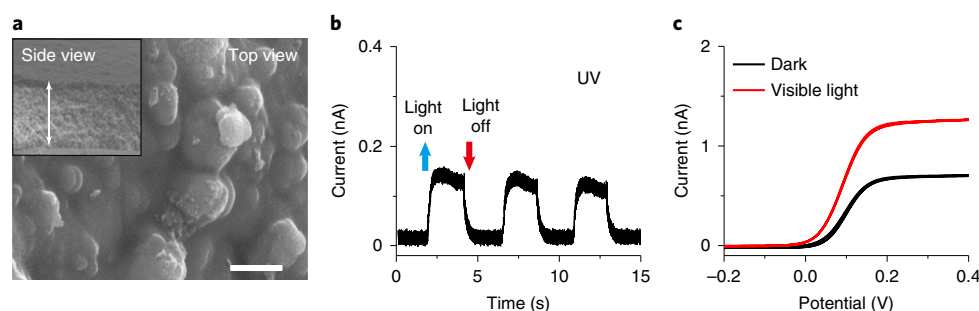


Fig. 4 | Characterization of TiO_2 @Au UMEs. **a**, SEM images of a nanoparticulate TiO_2 film on an Au UME. Scale bars, 200 nm (main); 4.4 μm (inset). **b**, Photocurrent response of TiO_2 @Au UME in DMSO solution containing 2.2 M water and 50 mM TBAP under UV light. **c**, Cyclic voltammograms of a TiO_2 @Au UME in a 0.1 M TEAPF_6 -acetonitrile solution containing 1 mM ferrocene and 1 mM N719 under visible light (red) and in the dark (black). Adapted with permission from ref. ⁴¹, American Chemical Society.

- 10 Calcine the TiO_2 -filmed Au UME by heating it. Program the temperature to increase to 500 °C at a rate of 10 °C/min and maintain the Au UME isothermally at 500 °C for 30 min. Allow the UME to cool to room temperature naturally for 3 h.

? TROUBLESHOOTING

- 11 To enhance the connectivity between the TiO_2 nanoparticles, immerse the TiO_2 @Au UME in 5 mL of 40 mM TiCl_4 aqueous solution at 70 °C for 30 min. Rinse the TiO_2 @Au UME with copious amounts of ethanol and Milli-Q water, and then let it dry under a nitrogen flow. Repeat Step 10.
- 12 Insert a thick copper wire (1 mm in diameter) into the capillary to establish an electrical connection with the TiO_2 @Au UME or UMEs for the electrochemical measurements. All procedures for the preparation of TiO_2 @Au UMEs are shown in Fig. 3.

■ PAUSE POINT To avoid the degradation of the TiO_2 film while in the UV environment or humid air atmosphere, store the fabricated TiO_2 @Au UMEs in a nitrogen-filled container in the dark at room temperature for up to 1 month.

- 13 Characterize the fabricated TiO_2 @Au UMEs to determine the morphology and photoconductivity of the TiO_2 film with SEM results and photoelectrochemical and electrochemical measurements (Fig. 4) as described in Box 1.

Experimental setup for photoelectrochemical measurement ● Timing ~24 h

- 14 Cut a silver wire (diameter, 1.0 mm) to a length of ~5 cm. Polish the wire with polishing cloths to remove its surface-oxide layer. Weld it to a copper wire (1 mm in diameter).

Box 1 | Characterization of TiO₂@Au UMEs ● Timing ~90 min**Procedure**

1 TiO₂@Au UME was used as the electron-transport electrode. Here, we provide a short protocol for the characterization of photoconductivity and electron mobility of nanoparticulate TiO₂ film, and we describe in detail additional characterization steps using SEM (option A), photoelectrochemical (option B) and electrochemical (option C) methods.

(A) Characterization of TiO₂@Au UMEs using SEM ● Timing ~40 min

- (i) Cut the tip of the TiO₂@Au UME (5 mm in length) with a grinding stone carefully.
- (ii) Attach the tip to the specimen holder, using three pieces of double-sided carbon tape.
▲ CRITICAL STEP Fix the tip with another piece of carbon tape to make sure its position does not change under vacuum.
- (iii) Insert the tip sample into the FE-SEM specimen chamber. Pump down and turn on high voltage to get an electron beam between 7 and 10 kV to avoid specimen charging and damage.
▲ CRITICAL STEP Coat the UME tip with Au using a benchtop sputter-coating system for 30 s before SEM measurement.
- (iv) Set the contrast and brightness to achieve the optimal image contrast. Scan at low magnification (×20) to find the surface of the TiO₂@Au UME and then increase the magnification to image selected surface morphology regions.
▲ CRITICAL STEP The parameters used are highly dependent on the SEM instrument.
- (v) Record the top-view image with a slow scan speed.

? TROUBLESHOOTING

- (vi) After tilting the specimen stage at the proper angle (~15°), repeat Step 1A(iv, v) to observe cross-sectional SEM images of the TiO₂@Au UMEs.

(B) Characterization of TiO₂@Au UMEs using photoelectrochemistry ● Timing ~25 min

- (i) Add 5 mL of DMSO solution containing 2.2 M water and 50 mM TBAP to a three-electrode glass electrochemical cell (Fig. 2, top left).
- (ii) Purge the electrolyte solution thoroughly by bubbling nitrogen for 10 min through the solution with a Teflon tube. A TiO₂@Au UME working electrode, a platinum wire counter electrode and an Ag/AgCl quasi-reference electrode are inserted into the solution of the electrochemical cell.
- (iii) Irradiate the tip surface of the TiO₂@Au UME electrode in the cell with light from a xenon lamp with an infrared light filter, through a light chopper with the shutter chopping frequency.
▲ CRITICAL STEP The electrode surface must be oriented toward the incident light.
- (iv) Collect amperometric *I*-*t* curves. Record the photocurrent response of the TiO₂@Au UME using a CHI660E electrochemical workstation with the following parameters: initial potential of +0.5 V; sample interval of 0.001 s; run time of 400 s and sensitivity of 1×10^{-9} A/V. Control the UV light (on and off) at a fixed chopping frequency of 0.4 Hz during the measurement process. The unit of the photocurrent is amperes.

? TROUBLESHOOTING**(C) Characterization of TiO₂@Au UMEs using electrochemistry ● Timing ~25 min**

- (i) Add 5 mL of 1 mM ferrocene and 1 mM N719 acetonitrile solution containing 0.1 M TEAPF₆ electrolyte to a three-electrode glass electrochemical cell (Fig. 2, top left). Repeat Step 1B(ii, iii).
- (ii) Perform cyclic voltammetry measurements using a CHI660E electrochemical workstation with the following parameters: potential range from -0.2 to +0.4 V; scan rate of 10 mV/s; sample interval of 0.001 s; quiet time of 2 s and sensitivity of 1×10^{-9} A/V.

- 15 Immerse the silver wire obtained from Step 14 and a platinum wire electrode in an electrochemical cell containing 10% (vol/vol) HCl solution. Apply an electric current of 0.15 μ A for 24 h to prepare the Ag/AgCl wire quasi-reference electrodes.

■ PAUSE POINT Prepare a batch of Ag/AgCl wire reference electrodes at one time to save time. Store in the dark under nitrogen at room temperature for up to 2 months.

- 16 Clean the electrochemical cell by submerging it in hot piranha solution in an open glass container for 10 min. Then sonicate the electrochemical cell in Milli-Q water for 10 min for three times to remove any remaining residues. Dry the cell in the oven.

! CAUTION Preparation and use of piranha solution should be carried out in a fume hood with proper personal protective equipment because it is corrosive and hazardous.

Photoelectrochemical measurement of an individual entity ● Timing ~1 h

- 17 Turn on the Axopatch 200B electrochemical instrument coupled to a xenon lamp. Set the low-pass filter to 5 kHz and the sampling rate to 100 kHz.

- 18 Focus the light from a xenon lamp on the surface of TiO₂@Au UME with the plano-convex lens and reflector. UV light with a wavelength <450 nm is filtered out using a filter.

▲ CRITICAL STEP Adjust the distance between the reflecting mirror and the TiO₂@Au UME to converge the light on a focal point on the TiO₂@Au UME surface. Block the light before the collision experiment to prevent the TiO₂ film and dye-tagged nanoparticles from deactivation.

- 19 Below we provide procedures for two separate experiments that use this ultrasensitive photoelectrochemical platform for single-entity collision detection. Use option A for photoelectrochemical measurement of a single N719@TiO₂ entity for a dye-sensitized solar cell system or option B for photoelectrochemical measurement of a single N719@ZnO entity for a water-splitting system.

▲ CRITICAL Perform all single-entity photoelectrochemical measurements in a Faraday cage at 25 °C.

(A) Photoelectrochemical measurement of a single N719@TiO₂ entity for dye-sensitized solar cell system

- (i) Add 1.8 mL of acetonitrile and 20 µL of I[−]/I₃[−] stock electrolyte solution to a photoelectrochemical cell and bubble argon for 10 min to remove oxygen dissolved in the solution.

- (ii) Connect a TiO₂@Au UME and an Ag/AgCl electrode to the headstage of the Axopatch 200B patch clamp system. Immerse the two electrodes in the electrolyte solution in the electrochemical cell.

▲ CRITICAL STEP Maintain the photoelectrochemical cell under an argon atmosphere throughout the experiment.

- (iii) Apply a positive potential (+600 mV versus Ag/AgCl) to the TiO₂@Au UME. Record current–time (*I*–*t*) curves in the dark and under visible light.

▲ CRITICAL STEP The applied potential used in the acquisition of the chronoamperometric curves should be more positive than the photocurrent onset potential of the N719 molecule (enabling electron injection from the N719 into the TiO₂ film), but more negative than its oxidation potential (Supplementary Fig. 3). In addition, the optimal external bias of +600 mV versus Ag/AgCl at the TiO₂@Au UME should be used for the collision experiments (Supplementary Fig. 4).

- (iv) Filter the N719@TiO₂ suspension solution through a hydrophobic PTFE membrane filter (0.22 µm) and add 200 µL of the solution to the photoelectrochemical cell to a final concentration of 174 pM.

▲ CRITICAL STEP To avoid the agglomeration of N719@TiO₂ entities, further sonicate the solution after filtering.

- (v) Record the photoelectrochemical current responses of individual N719@TiO₂ entities under visible light. Block the light to confirm the current transients result from the photoinduced electron transfer of N719@TiO₂ entities (Fig. 5).

(B) Photoelectrochemical measurement of a single N719@ZnO entity for a water-splitting system

- (i) Add 2 mL of DMSO containing 50 mM TBAP and 2.2 M water to a photoelectrochemical cell and bubble argon for 10 min to remove oxygen dissolved in the solution.

- (ii) Repeat Step 19A(ii–v). Note that N719@TiO₂ nanoparticles and I[−]/I₃[−] electrolyte solution should be replaced with N719@ZnO nanoparticles and a DMSO–water solution, respectively, in this photoelectrochemical measurement.

- (iii) Rinse the photoelectrochemical cell using ethanol and water. Then clean the electrochemical cell as described in Step 16.

- (iv) Investigate the photoelectrochemical behaviors of individual N719@ZnO entities as a function of TiO₂ film thicknesses. TiO₂@Au UMEs with different TiO₂ film thickness (3.1, 4.9, 5.8, 7.5, 8.6 and 9.4 µm) are made and used in each experiment according to Step 19B(i and ii) (Fig. 6).

? TROUBLESHOOTING

Data processing using custom-designed software and OriginLab ● Timing ~1.5 h

- 20 Open the custom-designed data analysis software based on a second-order differential-based calibration method, load the recorded photoelectrochemical data and set the analysis parameters, including the start and end times, the noise level, the mean current baseline, the margin of error and the threshold of interval time. The noise and baseline level can be visualized with the ‘trajectory viewer’ panel.

▲ CRITICAL STEP Set the current signal to at least triple the noise for the data analysis. The mathematical expressions used to process data for the current signals analysis and the source code of this custom-made data analysis software are provided in Supplementary Data 1 and 2,

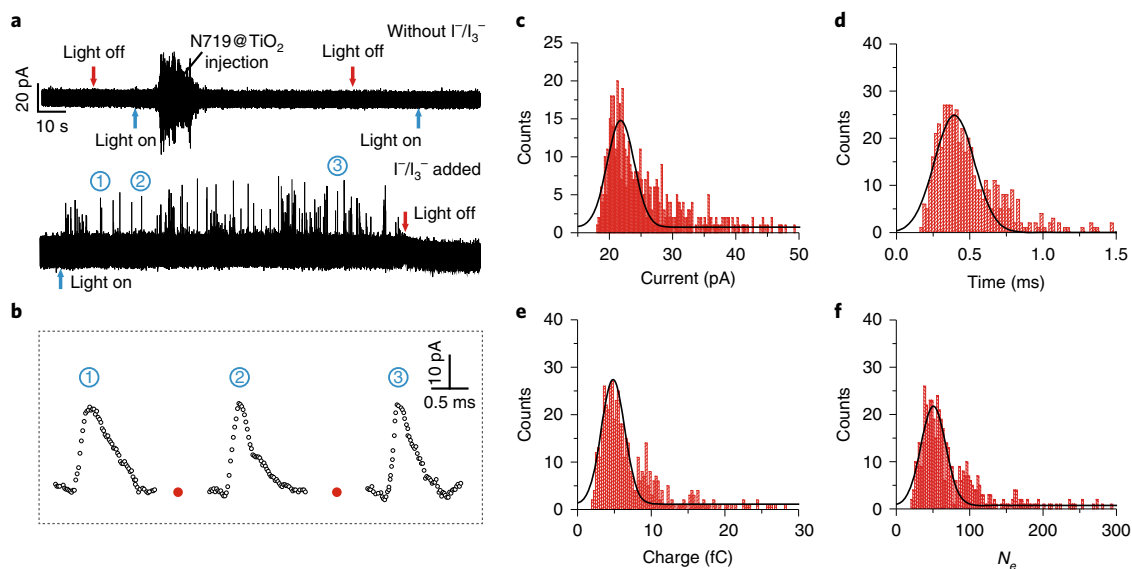


Fig. 5 | Photoelectrochemical response of a single N719@TiO₂ entity for a dye-sensitized solar cell system. **a**, Current-time curves of individual N719@TiO₂ entities colliding at a TiO₂@Au UME at +600 mV versus Ag/AgCl in the absence (top) and presence (bottom) of I⁻/I₃⁻ electrolyte solution under visible light. **b**, Expanded view of three representative time-resolved current traces from **a**. **c-f**, Histograms of the transient current (**c**), the duration (**d**), the integrated charge (**e**) for individual N719@TiO₂ nanoparticle events and the effective collected electron number (N_e, **f**) of a single N719 molecule per a typical collision. The number of collision events in each histogram is >1,000. The individual N719@TiO₂ nanoparticles afforded asymmetrical current traces with an amplitude of 21.5 ± 2.0 pA, a duration of 0.4 ± 0.1 ms and an integrated charge of 4.9 ± 1.3 fC. Thus, the corresponding number of collected electrons on an Au UME is 51 ± 4 per N719 molecule for a typical collision. Adapted with permission from ref. ⁴⁰, Wiley.

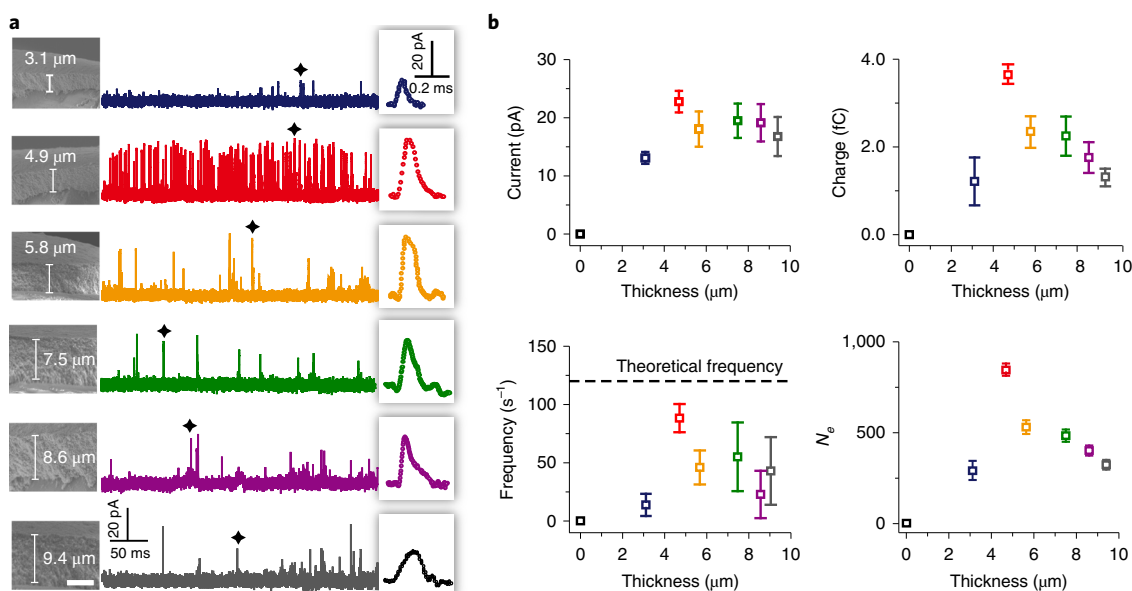


Fig. 6 | Photoelectrochemical responses of a single N719@ZnO entity for visible light water splitting. **a**, Current-time curves for individual N719@ZnO entities at TiO₂@Au UMEs of different TiO₂ film thicknesses ranging from 3.1 to 9.4 μm. SEM images of the coated TiO₂ film (left) and expanded views of the representative photocurrent transients that are marked by the shuriken symbols in the current-time curves (right). **b**, Plots of the variation of the current magnitude (top left), the integrated charge (top right), the collision frequency (bottom left) and N_e as a function of TiO₂ film thickness (bottom right). Error bars represent the standard deviation of three independent experiments. Individual current traces were observed for N719@ZnO entities at TiO₂@Au UMEs of different thicknesses, which are due to the varying diffusion and recombination processes of the photoinjected electrons in the different nanoparticulate TiO₂ films. The colors of the points in **b** correspond to those of the current-time curves in **a** and represent measurements of the associated entities. Adapted with permission from ref. ⁴¹, American Chemical Society.

respectively. For more user information about this software, see the user guide in the Supplementary Manual.

- 21 Click the 'Start' analysis button for data processing. The loaded data is automatically processed to obtain photoelectrochemical information about individual events in regard to current magnitude (I), duration time (T) and integrated charge (Q).
▲ CRITICAL STEP Check the 'event fits' panel to check for processing mistakes due to inappropriate parameters. Moreover, to minimize errors in the analysis data, event signals with a short duration ($<40\ \mu\text{s}$) at a bandwidth of 5 kHz should be excluded.
- 22 Export the database as a XLSX-format file after the completion of data processing and analysis. Open the target contents in OriginLab for the next analysis. Typically, work out the histograms (I , T and Q) as well as the corresponding distribution by a single Gaussian fitting (mean and standard deviation).
▲ CRITICAL STEP Set an optimal sampling bin size for Gaussian fitting with a coefficient of determination $R^2 > 0.9$.

Confirmation of a single-entity photoelectrochemical event ● Timing ~4 h

- 23 To ensure the photoelectrochemical events occurred at a genuinely single-entity level, with consistent measurement and reporting between laboratories, it is necessary to keep the suspension solution monodispersed throughout the entire measurement process. Three types of methods are used to measure size distribution of the suspended-entity solution before and after the photoelectrochemical experiment in this protocol. The physical and morphological characterizations are carried out by DLS (option A) and TEM (option B), respectively. We also correlate the entity concentration with collision frequency and confirm that it follows Fick's diffusion law (option C).

(A) DLS determination of size (hydrodynamic diameter) distribution of a single-collision event ● Timing ~30 min

- (i) Power on the Zetasizer instrument 30 min before use. Turn on the computer and open the Zetasizer software.
▲ CRITICAL STEP The following steps describe size characterization using a Zetasizer Nanoseries DLS instrument. For other DLS instruments, refer to the manufacturer's instructions.
- (ii) Pipette 1.0 mL of dye-tagged semiconductor nanoparticle suspension obtained in Box 2, before and after a photoelectrochemical experiment, into two Malvern disposable cuvettes. Insert one cuvette into the sample compartment of the Zetasizer instrument and close the lid.
▲ CRITICAL STEP Filter the sample with a PTFE membrane filter ($0.22\ \mu\text{m}$) before DLS measurements.
▲ CRITICAL STEP The following step should be applied to both cuvettes, one after the other.
- (iii) Select a material-medium-specific standard operating procedure to collect DLS data. See Supplementary Fig. 5.

? TROUBLESHOOTING

(B) TEM characterization of entity solution ● Timing ~1 h

- (i) Deposit 10 μL of the aliquot of dye-tagged semiconductor nanoparticle suspension from step 1A(v) or 1B(vii) of Box 2 onto a carbon-coated copper TEM grid and let it dry naturally for 40 min.
- (ii) Insert the sample into the JEOL JEM-2100F TEM specimen chamber and pump down.
- (iii) Set the contrast and brightness to obtain the optimum image contrast. Find individual entities in the sample at a low magnification ($\times 15\ \text{k}$ for the JEOL JEM-2100F TEM), spread the electron beam to its widest, increase the magnification ($\times 600\ \text{k}$ for the JEOL JEM-2100F TEM), and focus the electron beam slowly.
▲ CRITICAL STEP The parameters used are highly dependent on the SEM instrument.
? TROUBLESHOOTING
- (iv) Record the high-resolution TEM image at an exposure time of 0.2 s. TEM images of N719@ZnO entities are provided in Supplementary Fig. 6.
▲ CRITICAL STEP Avoid exposure of the sample to electron beam irradiation for $>2\ \text{min}$ in order to obtain high-quality and high-resolution TEM images.

Box 2 | Modification and characterization of dye-tagged oxide semiconductor entities**● Timing ~18 h****Procedure**

1 To modify and characterize N719@TiO₂ entities, use option A; to modify and characterize N719@ZnO entities, use option B.

(A) Modification and characterization of N719@TiO₂ entities

- (i) Dissolve 0.0137 g of 25-nm TiO₂ nanoparticle powder in 20 mL of methanol in a glass vial and then sonicate the suspension solution in an ultrasonic cleaner.
 - (ii) Dissolve 0.0110 g of N719 dye in 50 mL of acetonitrile in a glass vial.
 - (iii) Slowly add 10 mL of the methanol solution obtained in step 1A(i) to 10 mL of the acetonitrile solution obtained in step 1A(ii) and mix by vigorous vortexing at 2,500 r.p.m. for 5 min. Stir the resultant solution at 200 r.p.m. for 12 h continuously.
 - (iv) Centrifuge at 10,621g for 8 min (25 °C). Wash with acetonitrile and centrifuge three times (10,621g, 8 min, 25 °C) to remove excess N719.
 - (v) Resuspend and dilute the N719@TiO₂ nanoparticles in acetonitrile to an optimal concentration of 174 nM.
- ▲ CRITICAL STEP** To avoid aggregates of N719@TiO₂ nanoparticles, single-entity photoelectrochemical detection should be carried out as soon as the modification is finished.
- (vi) Record UV-visible spectra of N719 and the prepared N719@TiO₂ suspension, following the instrument's operating procedures. Further details can be found in Supplementary Fig. 11a.

? TROUBLESHOOTING

- (vii) Obtain FTIR spectra of TiO₂ nanoparticles, N719 solution and the N719@TiO₂ entity suspension. Follow your instrument's operating procedures. Supplementary Fig. 11b shows typical FTIR spectra for these three materials.

? TROUBLESHOOTING

- (viii) Acquire TEM images of N719@TiO₂ nanoparticles. Pipette ~10 µL of N719@TiO₂ suspension onto a carbon-coated copper TEM grid and allow the solution to dry. Follow your instrument's operating procedures to image the sample (typical conditions: 80 kV). Refer to Supplementary Fig. 11c for a representative TEM image.
- (ix) Collect DLS data for the N719@TiO₂ entity solution by following Step 23A. Typical data are shown in Supplementary Fig. 11d.

(B) Modification and characterization of N719@ZnO entities

- (i) Dissolve 2.95 g of Zn(OAc)₂·2H₂O and 1.48 g of KOH in 125 mL of methanol and 65 mL of methanol, respectively, in glass bottles.
 - (ii) Add the KOH solution to the Zn(OAc)₂ solution dropwise at 60 °C over 10 min with vigorous stirring at 2,500 r.p.m.
 - (iii) Stir this solution at 200 r.p.m. for 2 h at 60 °C, and then let it precipitate for an additional 2 h.
 - (iv) Centrifuge the obtained solution at 10,621g for 8 min at 25 °C. Wash with methanol and centrifuge (10,621g, 25 °C, 8 min) twice more. Dry in a vacuum oven at 100 °C for 10 h and store in an enclosed vial under nitrogen.
- PAUSE POINT** The synthesized ZnO nanoparticles can be stored in a nitrogen-filled container in the dark at 20 °C for 2 months.
- (v) Disperse 8 mg of ZnO nanoparticle powder in 10 mL of chloroform in a glass vial. Sonicate the obtained cloudy solution in a ultrasonic cleaner for 10 min until it becomes clear.
 - (vi) Dissolve 0.0181 g of N719 in 5 mL of DMF in a glass vial.
 - (vii) Repeat step 1A(iii–viii). Note that TiO₂ nanoparticles and acetonitrile solution should be replaced with ZnO nanoparticles and DMSO solution, respectively. Characterization data for N719@ZnO entities can be found in Supplementary Fig. 12.

(C) Dependence of collision frequency on entity concentration ● Timing ~3 h

- (i) Perform single-entity photoelectrochemical experiments with different concentrations (100, 200, 280 and 372 pM) of entity solution, following Step 19 of the main Procedure.
- ▲ CRITICAL STEP** Sonicate the suspension solution for 5–10 min before photoelectrochemical measurements.
- (ii) Analyze all the data files and count the collision frequencies of individual events at different concentrations. Plot experimental collision frequencies as a function of the entity concentration.
 - (iii) Investigate the correlation between the theoretical collision frequency (f_{NP}) and the nanoparticle concentration (C_{NP}) given by $f_{NP} = 4D_{NP}C_{NP}r_{UME}N_A$, where D_{NP} is the diffusion coefficient of nanoparticles, N_A is Avogadro's constant and r_{UME} is the radius of the TiO₂@Au UME. The value of D_{NP} can be estimated using the Stokes–Einstein relationship by $D_{NP} = k_B T / (6\pi\eta r_{NP})$, where k_B is Boltzmann's constant, T is temperature, η is the viscosity of the solution and r_{NP} is the radius of the N719@TiO₂ or N719@ZnO nanoparticle. The experimental results and the theoretical ones (calculated according to Fick's diffusion laws) are plotted in Supplementary Fig. 7.

Troubleshooting

Troubleshooting advice can be found in Table 1.

Table 1 | Troubleshooting table

Step	Problem	Possible reason	Solution
3	The pulled capillaries do not break cleanly	The heater level is insufficient to melt the capillary or the pulling force is weak	Gradually raise the heater level or increase the pulling force
5	The Au wire is stuck on the inner wall of the capillary and does not reach to the tip	The electrostatic interaction between the Au wire and the capillary is too strong	Wear an anti-static bracelet during the operation
7	Bubbles are generated while sealing the micropipette tip with Au wire	The vacuum is not high enough The heater level is too high	Increase the pumping time to achieve a better vacuum before sealing Lower the heater level
10	The TiO ₂ film cracks on the surface of Au UME	The program temperature rises too quickly	Lower the rate of increase of the program temperature
19B(iv)	There is too much noise during the electrochemical measurements	Grounding problem with the Faraday cage or electrochemical instrument	Adjust the grounds
23A(iii)	Multimodal size distribution in DLS measurement	Nanoparticles will start to aggregate if measurements take too long	Prepare fresh solution each time and always sonicate the solution before each experiment
23B(iii)	Charging of specimen in TEM	There is not enough carbon coating or copper tape on the coverslips	Remove the sample and try again with more tape or carbon-coat the surface again
Box 1, 1A(v)	The TiO ₂ film cracks or falls off the surface of the Au UME	The cooling process is too quick	Leave the electrodes to cool naturally to room temperature in Step 9
Box 1, 1B(iv)	No photocurrent response for the TiO ₂ @Au UME	The electrical connection of copper wire to the Au UME is not good The TiO ₂ film is damaged or falls off the tip	Reconnect a copper wire using a conductive Ag epoxy Redo the preparation of the TiO ₂ @Au UME, following Steps 9–11
Box 2, 1A (vi, vii)	Unexpected peaks are found on the UV-visible and FTIR spectra	The sample is contaminated	Preserve the samples in inert atmosphere (N ₂ or Ar) until the spectra analysis

Timing

- Steps 1–13, fabrication and characterization of TiO₂@Au UMEs: ~24 h
- Steps 14–16, experimental setup for photoelectrochemical measurement: ~24 h
- Steps 17–19, photoelectrochemical measurement of an individual entity: ~1 h
- Steps 20–22, data processing using custom-designed software and OriginLab: ~1.5 h
- Steps 23, confirmation of a single-entity photoelectrochemical event: ~4 h

Anticipated results

In this protocol, we have described detailed procedures for fabrication of an ultrasensitive photoelectrochemical platform, which could be widely used for real-time monitoring of the electron-transfer process of a single entity in various photoelectrochemical systems. Two types of oxide semiconductor (TiO₂ and ZnO) nanoparticles with N719 dye modification were prepared for intrinsic photoelectrochemical behavior measurements in dye-sensitized solar cells and visible-light water-splitting systems, respectively. UV-visible and FTIR spectra (Step A(vi, vii) and step 1B(vii) of Box 2) were used to characterize N719 molecules anchored on the surface of ZnO or TiO₂ nanoparticles via bidentate carboxylate linkages (Supplementary Figs. 11 and 12). For TiO₂@Au UME, a cross-sectional SEM image shows a compact TiO₂ film with a thickness of 4.4 μm (Fig. 4a). Top-view SEM images of TiO₂@Au UMEs reveal that the desired highly accessible nanocrystalline TiO₂ nanoparticulate film has been coated onto the surface of the Au UME. Figure 4b displays the time-resolved photocurrent responses of TiO₂@Au UME as the UV light is turned on and off. In addition, Fig. 4c shows a typical UME voltammogram with a well-resolved limiting current for 1-mM ferrocene, 1-mM N719 acetonitrile solution at the TiO₂@Au UME surface. An enhanced limiting current

was observed under visible-light illumination as compared to the data obtained in dark conditions. All characterization results demonstrated the successful fabrication of a micron-thickness nanoparticulate TiO_2 film with a substantially high electron mobility and photoconductivity.

Current–time curves of individual N719@TiO_2 entities colliding with a $\text{TiO}_2\text{@Au}$ UME are collected in acetonitrile solutions containing 10 mM LiI and 1 mM I_2 (Step 19A). Under visible light, substantial current transients were observed at +600 mV after the injection of the freshly prepared N719@TiO_2 nanoparticles into the electrolyte solution; however, negligible transient signals were obtained in the dark (Fig. 5a). This indicates the photoinduced electron transfer does not occur without photoexcitation. Note that no current signal was observed for the neat electrolyte solution without N719@TiO_2 entities, with illumination or without. Moreover, current–time responses showed a flat curve (no current spikes) in the absence of I^-/I_3^- redox electrolyte under illumination. This is because the I^-/I_3^- redox pair acts as an efficient electron donor for N719. There would be no continuing electron flow in this case if there were no I^-/I_3^- present in the electrolyte. In addition, no current transient was observed for individual bare TiO_2 nanoparticles without the N719 dye modification in an amperometric current–time curve (Supplementary Fig. 8), demonstrating that the collected electrons result from the photo-excited N719 moieties under visible-light illumination. Figure 5b shows the representative current traces that initially rise quickly and decay at a slower rate for individual N719@TiO_2 entities. The most population of collision events had a mean duration of 0.4 ± 0.1 ms (Fig. 5c) and a mean amplitude of 21.5 ± 2.0 pA (Fig. 5d). Considering the electron recombination in a multiple-trapping TiO_2 film, the effective number of collected electrons for a single N719 molecule per typical collision (N_e) at a $\text{TiO}_2\text{@Au}$ UME from the measured transients was calculated by $N_e = Q/neN_{\text{N719}}$, where Q is the integrated charge of individual events, n is the number of electrons involved in the photoelectrochemical reaction (1 for N719), e is the elementary charge and N_{N719} is the average number of N719 molecules modified onto the surface of a single TiO_2 nanoparticle surface (~ 600 ; Supplementary Fig. 9). Average integrated charge of individual events was estimated to be 4.9 ± 1.3 fC from photocurrent transients (Fig. 5e). This allows us to calculate an N_e value of 51 ± 4 per N719 molecule (Fig. 5f), revealing the intrinsic electron-transfer number of a single N719 molecule for a typical collision. In this measurement, the TiO_2 film is essential to the successful trapping of the photoinduced electron-transfer process because no collision event could be observed for N719@TiO_2 entities at a bare Au UME (Supplementary Fig. 1). This could be attributed to instantaneous collisions of individual entities beyond the measurement capability of current electrochemical instrumentation or high energy dissipation for the transfer of electrons through the TiO_2 semiconductor–Au metal interface⁵⁶.

As a parallel study, we also investigated single-entity photoelectrochemistry for N719@ZnO nanoparticles suspended in a DMSO–water solution at $\text{TiO}_2\text{@Au}$ UMEs with different TiO_2 film thicknesses (Step 19B). Owing to the varying diffusion and recombination processes of the photo-injected electrons in the different nanoparticulate TiO_2 networks, thickness-dependent current traces were observed for individual N719@ZnO nanoparticle collisions corresponding to single-entity photocatalytic water oxidation (Fig. 6a). The current, charge, collision frequency and N_e of individual N719@ZnO nanoparticles as functions of the TiO_2 film thickness were plotted (Fig. 6b). A representative current–time curve at a 4.4- μm -thick $\text{TiO}_2\text{@Au}$ UME under visible light or in the dark is shown in Supplementary Fig. 10a. The photocatalytic water oxidation for oxygen evolution was further demonstrated by gas chromatographic results under visible-light irradiation (Supplementary Fig. 10b)⁴¹. Furthermore, the turnover number ($\text{TON} = Q/neN_{\text{N719}}$) for oxygen evolution of a single N719 per collision was estimated to be 285 ± 12 at a 4.4- μm $\text{TiO}_2\text{@Au}$ UME (Supplementary Fig. 10c). In addition, the measured results of TiO_2 film thickness-dependent current traces further demonstrated that a trap-limited electron diffusion process was the rate-limiting step in our single-entity photoelectrochemical measurement.

Reporting Summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

References

1. Danaher, W. J. & Lyons, L. E. Photoelectrochemical cell with cadmium telluride film. *Nature* **271**, 139 (1978).
2. Grätzel, M. Photoelectrochemical cells. *Nature* **414**, 338–344 (2001).

3. Youngblood, W. J. et al. Photoassisted overall water splitting in a visible light-absorbing dye-sensitized photoelectrochemical cell. *J. Am. Chem. Soc.* **131**, 926–927 (2009).
4. Zewdu, T., Clifford, J. N., Hernández, J. P. & Palomares, E. Photoinduced charge transfer dynamics in efficient TiO₂/CdS/CdSe sensitized solar cells. *Energy Environ. Sci.* **4**, 4633–4638 (2011).
5. Tvrdy, K., Frantsuzov, P. A. & Kamat, P. V. Photoinduced electron transfer from semiconductor quantum dots to metal oxide nanoparticles. *Proc. Natl Acad. Sci. USA* **108**, 29–34 (2011).
6. Murphy, C. et al. Long-range photoinduced electron transfer through a DNA helix. *Science*. **262**, 1025–1029 (1993).
7. Lewis, F. D. et al. Distance-dependent electron transfer in DNA hairpins. *Science*. **277**, 673–676 (1997).
8. Bard, A. J. Photoelectrochemistry. *Science*. **207**, 139–144 (1980).
9. Joshi, U. A., Palasyuk, A., Arney, D. & Maggard, P. A. Semiconducting oxides to facilitate the conversion of solar energy to chemical fuels. *J. Phys. Chem. Lett.* **1**, 2719–2726 (2010).
10. Urbani, M., Grätzel, M., Nazeeruddin, M. K. & Torres, T. Meso-substituted porphyrins for dye-sensitized solar cells. *Chem. Rev.* **114**, 12330–12396 (2014).
11. Wu, J. H. et al. Electrolytes in dye-sensitized solar cells. *Chem. Rev.* **115**, 2136–2173 (2015).
12. Peña, M. A. & Fierro, J. L. G. Chemical structures and performance of perovskite oxides. *Chem. Rev.* **101**, 1981–2018 (2001).
13. Reier, T. et al. Molecular insight in structure and activity of highly efficient, low-Ir Ir-Ni oxide catalysts for electrochemical water splitting (OER). *J. Am. Chem. Soc.* **137**, 13031–13040 (2015).
14. Buurmans, I. L. C. & Weckhuysen, B. M. Heterogeneities of individual catalyst particles in space and time as monitored by spectroscopy. *Nat. Chem.* **4**, 873–886 (2012).
15. Ma, W. et al. Electrochemical size measurement and characterization of electrodeposited platinum nanoparticles at nanometer resolution with scanning electrochemical microscopy. *Nano Lett.* **17**, 4354–4358 (2017).
16. Su, Y. et al. Single-nanowire photoelectrochemistry. *Nat. Nanotechnol.* **11**, 609–612 (2016).
17. Fang, Y. et al. Intermittent photocatalytic activity of single CdS nanoparticles. *Proc. Natl Acad. Sci. USA* **114**, 10566–10571 (2017).
18. Quinn, B. M., van't Hof, P. G. & Lemay, S. G. Time-resolved electrochemical detection of discrete adsorption events. *J. Am. Chem. Soc.* **126**, 8360–8361 (2004).
19. Xiao, X. Y. & Bard, A. J. Observing single nanoparticle collisions at an ultramicroelectrode by electrocatalytic amplification. *J. Am. Chem. Soc.* **129**, 9610–9612 (2007).
20. Sardesai, N. P., Andreescu, D. & Andreescu, S. Electroanalytical evaluation of antioxidant activity of cerium oxide nanoparticles by nanoparticle collisions at microelectrodes. *J. Am. Chem. Soc.* **135**, 16770–16773 (2013).
21. Xiao, X. Y., Fan, F. R. F., Zhou, J. & Bard, A. J. Current transients in single nanoparticle collision events. *J. Am. Chem. Soc.* **130**, 16669–16677 (2008).
22. Dick, J. E. & Bard, A. J. Recognizing single collisions of PtCl₆²⁻ at femtomolar concentrations on ultramicroelectrodes by nucleating electrocatalytic clusters. *J. Am. Chem. Soc.* **137**, 13752–13755 (2015).
23. Zhao, L. J., Qian, R. C., Ma, W., Tian, H. & Long, Y.-T. Electrocatalytic efficiency analysis of catechol molecules for NADH oxidation during nanoparticle collision. *Anal. Chem.* **88**, 8375–8379 (2016).
24. Kim, B. K., Boika, A., Kim, J., Dick, J. E. & Bard, A. J. Characterizing emulsions by observation of single droplet collisions-attoliter electrochemical reactors. *J. Am. Chem. Soc.* **136**, 4849–4852 (2014).
25. Dick, J. E., Renault, C. & Bard, A. J. Observation of single-protein and DNA macromolecule collisions on ultramicroelectrodes. *J. Am. Chem. Soc.* **137**, 8376–8379 (2015).
26. Zhou, Y. G., Rees, N. V. & Compton, R. G. The electrochemical detection and characterization of silver nanoparticles in aqueous solution. *Angew. Chemie Int. Ed* **50**, 4219–4221 (2011).
27. Holt, L. R., Plowman, B. J., Young, N. P., Tschulik, K. & Compton, R. G. The electrochemical characterization of single core-shell nanoparticles. *Angew. Chem. Int. Ed.* **55**, 397–400 (2016).
28. Oja, S. M. et al. Observation of multipeak collision behavior during the electro-oxidation of single Ag nanoparticles. *J. Am. Chem. Soc.* **139**, 708–718 (2017).
29. Ustarroz, J., Kang, M., Bullions, E. & Unwin, P. R. Impact and oxidation of single silver nanoparticles at electrode surfaces: one shot versus multiple events. *Chem. Sci.* **8**, 1841–1853 (2017).
30. Ma, W. et al. Tracking motion trajectories of individual nanoparticles using time-resolved current traces. *Chem. Sci.* **8**, 1854–1861 (2017).
31. Hao, R., Fan, Y. & Zhang, B. Imaging dynamic collision and oxidation of single silver nanoparticles at the electrode/solution interface. *J. Am. Chem. Soc.* **139**, 12274–12282 (2017).
32. Robinson, D. A. et al. Collision dynamics during the electrooxidation of individual silver nanoparticles. *J. Am. Chem. Soc.* **139**, 16923–16931 (2017).
33. Ma, W., Ma, H., Yang, Z. Y. & Long, Y.-T. Single Ag nanoparticle electro-oxidation: potential-dependent current traces and potential-independent electron transfer kinetic. *J. Phys. Chem. Lett.* **9**, 1429–1433 (2018).
34. Dick, J. E., Hilterbrand, A. T., Strawsine, L. M., Upton, J. W. & Bard, A. J. Enzymatically enhanced collisions on ultramicroelectrodes for specific and rapid detection of individual viruses. *Proc. Natl Acad. Sci. USA* **113**, 6403–6408 (2016).
35. Zhou, Y. G., Rees, N. V. & Compton, R. G. The electrochemical detection of tagged nanoparticles via particle-electrode collisions: nanoelectroanalysis beyond immobilisation. *Chem. Commun.* **48**, 2510–2512 (2012).

36. Corkum, P. B. & Krausz, F. Attosecond science. *Nat. Phys.* **3**, 381–387 (2007).
37. Schnadt, J. et al. Experimental evidence for sub-3-fs charge transfer from an aromatic adsorbate to a semiconductor. *Nature* **418**, 620–623 (2002).
38. Huber, R., Moser, J. E., Grätzel, M. & Wachtveitl, J. Real-time observation of photoinduced adiabatic electron transfer in strongly coupled dye/semiconductor colloidal systems with a 6 fs time constant. *J. Phys. Chem. B* **106**, 6494–6499 (2002).
39. Huber, R., Spörlein, S., Moser, J. E., Grätzel, M. & Wachtveitl, J. The role of surface states in the ultrafast photoinduced electron transfer from sensitizing dye molecules to semiconductor colloids. *J. Phys. Chem. B* **104**, 8995–9003 (2000).
40. Peng, Y. Y., Ma, H., Ma, W., Long, Y.-T. & Tian, H. Single-nanoparticle photoelectrochemistry at a nanoparticulate TiO₂-filmed ultramicroelectrode. *Angew. Chemie Int. Ed.* **57**, 3758–3762 (2018).
41. Ma, H. et al. Quantifying visible-light-induced electron transfer properties of single dye-sensitized ZnO entity for water splitting. *J. Am. Chem. Soc.* **140**, 5272–5279 (2018).
42. Lasser, L. et al. Energy level alignment at titanium oxide-dye Interfaces: Implications for electron injection and light harvesting. *J. Phys. Chem. C* **119**, 9899–9909 (2015).
43. Soga, T. (ed.) *Nanostructured Materials for Solar Energy Conversion* (Elsevier, 2006).
44. Boschloo, G. & Hagfeldt, A. Characteristics of the iodide/triiodide redox mediator in dye-sensitized solar cells. *Acc. Chem. Res.* **42**, 1819–1826 (2009).
45. Han, L. et al. High-efficiency dye-sensitized solar cell with a novel co-adsorbent. *Energy Environ. Sci.* **5**, 6057–6060 (2012).
46. Yella, A. et al. Porphyrin-sensitized solar cells with cobalt (II/III)-based redox electrolyte exceed 12 percent efficiency. *Science* **334**, 629–634 (2011).
47. Mishra, A., Fischer, M. K. & Bäuerle, P. Metal-free organic dyes for dye-sensitized solar cells: from structure: property relationships to design rules. *Angew. Chem. Int. Ed.* **48**, 2474–2499 (2009).
48. Robinson, D. A., Edwards, M. A., Ren, H. & White, H. S. Effects of instrumental filters on electrochemical measurement of single-nanoparticle collision dynamics. *ChemElectroChem* **5**, 3059–3067 (2018).
49. Kanokkanchana, K., Saw, E. N. & Tschulik, K. Nano impact electrochemistry: effects of electronic filtering on peak height, duration and area. *ChemElectroChem* **5**, 3000–3005 (2018).
50. Kim, J., Kim, B. K., Cho, S. K. & Bard, A. J. Tunneling ultramicroelectrode: nanoelectrodes and nanoparticle collisions. *J. Am. Chem. Soc.* **136**, 8173–8176 (2014).
51. Hill, C. M., Kim, J. & Bard, A. J. Electrochemistry at a metal nanoparticle on a tunneling film: a steady-state model of current densities at a tunneling ultramicroelectrode. *J. Am. Chem. Soc.* **137**, 11321–11326 (2015).
52. Ostojic, N., Thorpe, J. H. & Crooks, R. M. Electron transfer facilitated by dendrimer-encapsulated Pt nanoparticles across ultrathin, insulating oxide films. *J. Am. Chem. Soc.* **138**, 6829–6837 (2016).
53. Loussaert, J. A., Fosdick, S. E. & Crooks, R. M. Electrochemical properties of metal-oxide-coated carbon electrodes prepared by atomic layer deposition. *Langmuir* **30**, 13707–13715 (2014).
54. Acharya, S., Lancaster, M. & Maldonado, S. Semiconductor ultramicroelectrodes: Platforms for studying charge-transfer processes at semiconductor/liquid interfaces. *Anal. Chem.* **90**, 12261–12269 (2018).
55. Bentley, C. L., Perry, D. & Unwin, P. R. Stability and placement of Ag/AgCl quasi-reference counter electrodes in confined electrochemical cells. *Anal. Chem.* **90**, 7700–7707 (2018).
56. Chen, J. et al. Tunneling interlayer for efficient transport of charges in metal oxide electrodes. *J. the Am. Chem. Soc.* **138**, 3183–3189 (2016).

Acknowledgements

This research was supported by the National Natural Science Foundation of China (grants 21775043 and 21421004), Shanghai Municipal Natural Science Foundation (19ZR1472100), the Program of Introducing Talents of Discipline to Universities (B16017) and the Innovation Program of the Shanghai Municipal Education Commission (2017-01-07-00-02-E00023).

Author contributions

W.M. and Y.-T.L. conceived and designed the research. H.M., Y.-Y.P. and W.M. performed all experiments and analyzed the data. W.M. and H.M. wrote the manuscript. W.M. and Y.-T.L. finalized the preparation of the manuscript. Y.-T.L. and H.T. designed and managed the project. All the authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41596-019-0197-8>.

Reprints and permissions information is available at www.nature.com/reprints.

Correspondence and requests for materials should be addressed to H.T. or Y.-T.L.

Publisher's note: Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Received: 8 June 2018; Accepted: 15 May 2019;
Published online: 7 August 2019

Related links**Key references using this protocol**

Peng, Y.-Y., Ma, H., Ma, W., Long, Y.-T. & Tian, H. *Angew. Chem. Int. Ed.* **57**, 3758–3762 (2018):

<https://doi.org/10.1002/anie.201710568>

Ma, H. et al. *J. Am. Chem. Soc.* **140**, 5272–5279 (2018): <https://pubs.acs.org/doi/abs/10.1021/jacs.8b01623>

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see [Authors & Referees](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☒ | ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

Clampex

Data analysis

Matlab and Originlab

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Single entity photoelectrochemical measurement
Research sample	Dye-tagged oxide semiconductor nanoparticles
Sampling strategy	Monodispersed dye-tagged oxide semiconductor nanoparticle suspensions
Data collection	Clampex software
Timing and spatial scale	single entity photoelectrochemical measurement takes less than 1 hour for each independent experiment to complete.
Data exclusions	Photoelectrochemical data were excluded if nanoparticle suspensions start to aggregate.
Reproducibility	To ensure the photoelectrochemical events occurred at genuinely single entity level, with consistent measurement and reporting between different experiments, it is necessary to keep the suspension solution monodispersed throughout the entire measurement process.
Randomization	This is not relevant to our study because we statistically analyze the data of individual events.
Blinding	Our statistical results follow a Gaussian distribution.
Did the study involve field work?	☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging